

6	(a)(i)	either wavelength is longer than threshold wavelength or frequency is below the threshold frequency or photon energy is less than work function	B1
	(a)(ii)	$hc / \lambda = hf_0 + E_k$ $(6.63 \times 10^{-34} \times 3.00 \times 10^8) / (240 \times 10^{-9}) = 6.63 \times 10^{-34}f_0 + 4.44 \times 10^{-19}$ $f_0 = 5.8 \times 10^{14} \text{ Hz}$	C1 A1
	(b)1.	photon energy larger so (maximum) kinetic energy is larger	B1
	(b)2.	fewer photons (per unit time) so (maximum) current is smaller	B1 B1

7

Section B

7	(a)	The direction of the oscillations or vibrations is the key difference between	B1
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